



# TUDOR ANDREI HĂLĂȘAG

I build things. From audio encryption tools and OOP transit simulations to embedded hardware with Arduino — I've been engineering solutions independently since before university. Experienced in team leadership and operations management from running shift operations in a high-volume service environment. Looking to apply that hands-on engineering mindset in a real industrial context.

## PROJECTS & PERSONAL DEVELOPMENT

- Audio File Encryption/Decryption** · *Python* — Built a file-level audio encryption/decryption tool using SHA-256 hashing — full implementation from scratch.
- Public Transport Ticketing System** · *C++* — Designed an OOP simulation of Bucharest's transit network (metro, tram, bus) with ticket types, passes, and expiry logic.
- Chess Constraint Solver** · *C++* — Implemented the N-Queens combinatorial problem and related constraint-satisfaction algorithms.
- CV Generator Tool** · *C#* — Developed a desktop utility for automated document generation with structured data input.
- Step-Down Transformer Design** · *Engineering* — Designed a functional step-down transformer from scratch, applying electromagnetic theory and component selection.
- Arduino & Embedded Systems Lab** · *Hardware* — Programmed microcontrollers and interfaced LCD displays, sensors, stepper motors, and speakers — hands-on embedded I/O and memory management.

## EDUCATION

- BSc - Faculty of Robotics and Industrial Engineering** 2022 - Present  
*Polytechnic University of Bucharest, Bucharest*
  - Control systems, robotics, and industrial automation — with hands-on lab projects.
  - Systems-thinking applied to real engineering problems across hardware and software disciplines.
- Mathematics - Computer Science** 2017 - 2021  
*"MIHAIL KOGĂLNICEANU" NATIONAL COLLEGE, Galați*
  - Final Exam: 8.76 / 10**
  - Intensive track in advanced mathematics and programming.
  - Practical C++ and Python from the earliest stages — applied in personal projects throughout.

## EMPLOYMENT HISTORY

- Barista & Operations Lead** Mar 2023 - Sept 2023  
*5 TO GO EUROPE HOLDING SRL Romania, Bucharest*
  - Led shift operations: team coordination, opening/closing procedures, and workflow management in a high-volume environment.
  - Managed all transactions and inventory via Comax Smart Retail (ERP/POS) and Excel — 100% cash-handling accuracy throughout.
  - Drove a 15–20% improvement in peak-hour service efficiency through process restructuring and preparation systems.

## Personal

- Name**  
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## Skills

- C / C++
- Python
- Microsoft Office Suite
- MATLAB
- Arduino
- AutoCAD Mechanical
- LabVIEW
- ORACLE SQL
- C# / Java
- Adobe Illustrator
- Tina-TI / LTspice

## Languages

|          |        |
|----------|--------|
| Romanian | Native |
| English  | C2     |
| Italian  | A2     |
| Turkish  | A2     |

## Interests

- Music (drums, guitar, bass, DJ)
- Sports — gym & outdoor fitness
- Travel & hiking trips